

Hemanth Chaparla | Data Engineer

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SUMMARY

Data Analyst / Data Engineer with 4+ years of experience improving data reliability, stabilizing pipelines, and delivering clean, analysis-ready datasets using SQL, Python, Spark, Databricks, and Delta Lake. Skilled in diagnosing data issues, aligning KPIs, and resolving inconsistencies across reporting layers. Known for reducing refresh failures, improving data quality, and being the go-to person when dashboards break or KPIs don't match. Open to junior-mid Data Analyst, Data Engineer, and Analytics Engineer roles focused on ETL workflows, data quality, and reliable analytics delivery.

TECHNICAL SKILLS

Programming: SQL, Python, R

Analytics & BI: Tableau, Power BI, KPI Development, Dashboarding, Forecasting, Cohort Analysis, Excel

Data Engineering: Spark, Databricks, Delta Lake, ETL/ELT Pipelines, Data Modeling, Partitioning, Incremental Loads

Cloud: AWS (Glue, S3, Lambda, EC2), Azure Data Factory

Databases: PostgreSQL, MySQL, Snowflake, MongoDB

ML (Exposure): Logistic Regression, XGBoost, Feature Engineering, Scikit-learn

Tools: Git, Docker, Jira, Confluence, VS Code, Jupyter Notebook

PROFESSIONAL EXPERIENCE

Databricks, USA

July 2025 – Present

Data Engineer

- Delivered Spark/SQL pipelines moving raw → bronze → silver layers, enabling daily reporting for 4+ analytics teams.
- Standardized Delta tables via schema alignment and incremental logic, reducing dashboard inconsistencies by 18.7% (per monthly defect logs).
- Implemented validation checks (row counts, schema drift, missing partitions), preventing 32 recurring data defects per quarter.
- Automated anomaly indicators detecting 9–14 unexpected data drops per month.
- Resolved job failures and KPI mismatches, reducing BI escalations by 26.4% within two quarters.
- Collaborated with BI & product teams to align KPI definitions across finance, operations, and leadership reporting.
- Documented lineage and pipeline logic, improving troubleshooting time for 6+ analysts and new engineers.

Tech: Spark, SQL, Python, Delta Lake, Databricks, AWS

UnitedHealth Group, USA

June 2024 – May 2025

Data Engineer

- Managed Spark/Delta pipelines refreshing 8–10M+ claims records daily, used by fraud, network, and operations teams.
- Optimized joins and pruning strategies, improving refresh runtime by 19.3% across 7 monitored workflows.
- Reduced manual reconciliation workload by 27% through SQL cleanup and transformation simplification.
- Built Databricks SQL monitoring views used by engineering managers for SLA checks and ingestion diagnostics.
- Implemented audit checks (duplicate scans, missing columns, null-pattern detection), reducing data issues by 38.6%.
- Collaborated with BI, product, and engineering to align KPI logic across 5+ reporting teams, eliminating metric conflicts.
- Authored runbooks and troubleshooting guides that lowered ETL resolution time by 23–28%, depending on pipeline complexity.

Tech: Spark, SQL, Python, Delta Lake, Databricks SQL, Azure Data Factory

Vivma Software Inc., India

Aug 2022 – July 2023

Data Analyst

- Built Tableau dashboards visualizing financial KPIs, credit risk exposure, and customer behavior.
- Automated SQL/Python workflows, saving 6–8 hours weekly of manual analyst workload.
- Conducted churn, cohort, and forecasting analyses for budgeting and retention strategies.
- Standardized KPI definitions across risk and finance teams, improving reporting consistency.
- Prepared feature sets and cleaned datasets used in churn prediction models.

Tech: SQL, Python, Tableau, Snowflake, Excel

Adons Softech, India

Jan 2021 – July 2022

Data Analyst / Business Analyst

- Built Power BI/Tableau dashboards covering revenue trends, sales performance, and operational KPIs.
- Created SQL/Python ETL scripts integrating CRM, finance, and transactional data.
- Performed variance analysis and forecasting supporting leadership decisions.
- Automated recurring reports with validation steps, reducing manual effort and errors.
- Worked with cross-functional teams to clarify data requirements and improve accuracy.

Tech: SQL, Python, Tableau, Power BI, Excel

PROJECTS

Healthcare Claims Analytics & Fraud Detection

- Built a Spark + Delta Lake pipeline (bronze → silver → curated) processing 5.4M+ synthetic claims, resolving schema inconsistencies across sources.
- Engineered 45+ behavior/cost features and added DQ checks (drift, null spikes), improving dataset stability by ~17%.
- Designed deviation scoring and anomaly detection logic surfacing potential fraud and provider outliers.
- Published curated datasets into Tableau/Power BI for claim cost and anomaly analysis.

Tech: SQL, Python, Spark, Delta Lake, Databricks, Tableau/Power BI

Customer Churn Prediction

- Built a Spark + Delta Lake churn pipeline integrating ~18–20M synthetic CRM/usage/support records with bronze → silver → gold layers.
- Engineered 140+ features (rolling windows, payment delay ratios, behavior metrics) with optimized PySpark transformations.
- Implemented drift and timestamp validation checks to maintain data consistency for modeling.
- Automated monthly churn scoring using Logistic Regression/XGBoost, improving early-risk detection by ~11%.

Tech: Databricks, Spark, Delta Lake, SQL, Python, XGBoost, Tableau

EDUCATION

Master of Science in Computer Science, Illinois Institute of Technology, Chicago, IL

Bachelor of Technology in Computer Science & Engineering, BML Munjal University, India

CERTIFICATION

- AWS Certified Data Engineer – Associate
- AWS Solutions Architect – Associate
- Snowflake SnowPro Core
- Tableau Desktop Specialist
- Databricks Generative AI Fundamentals